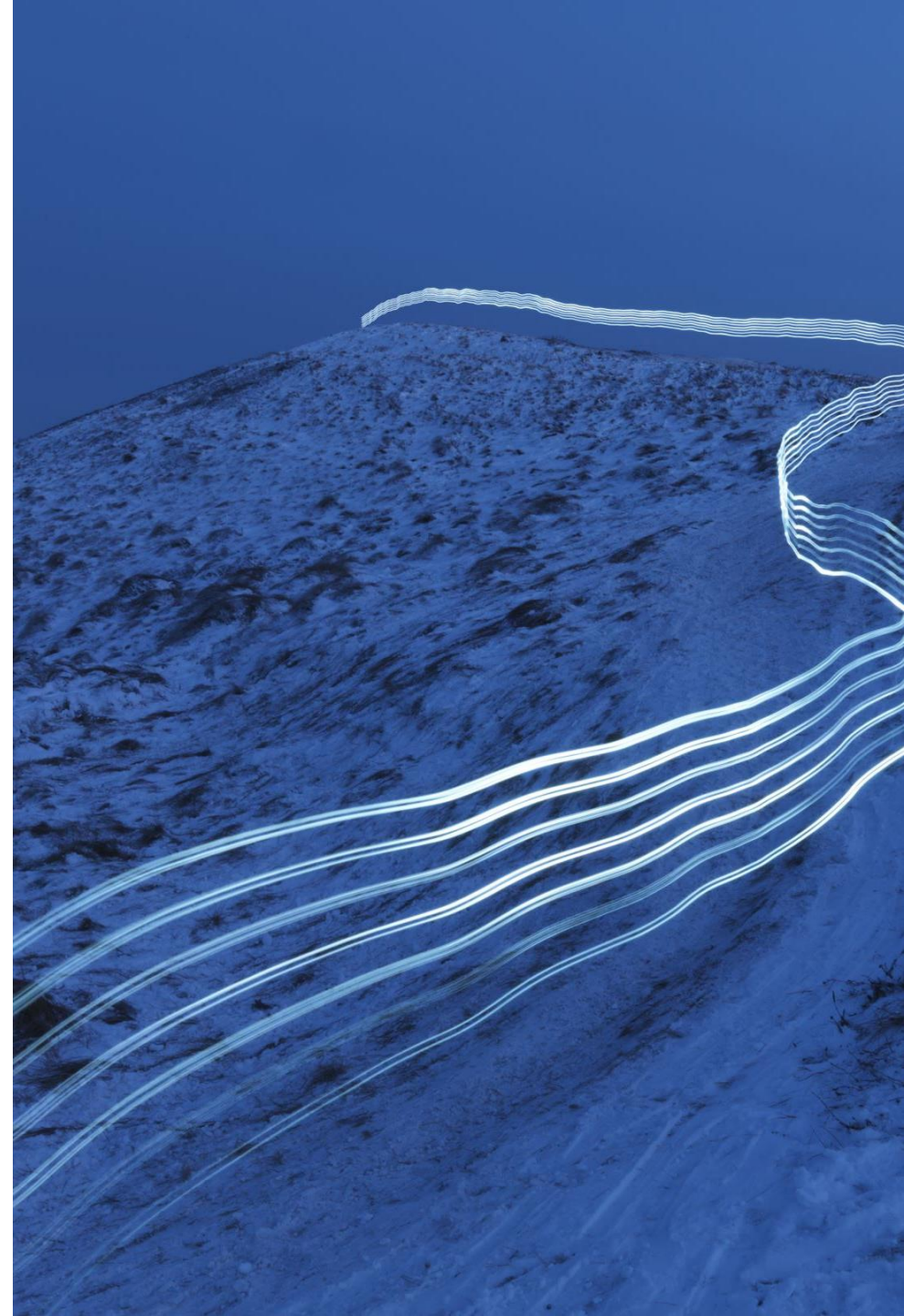


SCSI Pass-Through (SPT) UNMAPs

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Topics

- Overview of UNMAPs
- SCSI UNMAP Methods
- Host File System Support
- Futures

Overview of UNMAPs

- Supported on Thinly Provisioned Storage
- Allow arrays to efficiently reuse freed blocks
- Host File Systems support via File Trim Operations
 - File Trim operations converted into SCSI UNMAP commands
 - Windows enables File Trims automatically for NTFS
 - Linux file systems support this via the “**-o discard**” *mount* option
 - verified on ext4 and XFS file systems
 - [Veritas](#) supports similar via *fsadm* options
 - File Trims triggered by freeing blocks: file [delete](#), file [truncate](#)
- Host SCSI tools exist to verify UNMAP operations on direct disks

SCSI UNMAP Methods

- Multiple methods available:
 - SCSI UNMAP command (opcode **0x42**)
 - Write Same 16 with unmap (opcode **0x93,0x08**)
 - Write Using Token (WUT) with Zero ROD (**0x83,0x11**)
- SCSI UNMAP is the most popular method
- SCSI Read Capacity(16) reports provisioning information
- SCSI Get LBA Status reports mapped/deallocated blocks
- SCSI Inquiry VPD Block Limits reports maximum UNMAP information

Report Capacity & Provisioning Type

```
# export SCU_DEVICE="/dev/mapper/mpathu"  
# spt readcapacity16 logprefix=
```

Read Capacity(16) Data: (data length 32 bytes)

```
                Maximum Capacity: 409600000 (200000.000 megabytes)  
                    Block Length: 512  
                Protection Enabled: No  
                    Protection Type: 0  
    Logical Blocks per Physical Exponent: 0 (1 blocks per physical)  
        Protection Information Exponent: 0  
    Logical Block Provisioning Management: Yes (Thin Provisioned)  
    Logical Block Provisioning Read Zeroes: Yes (unmapped blocks read as zero)  
        Lowest Aligned Logical Block Address: 0  
#
```

Report UNMAP Information

```
# spt inquiry page=block_limits logprefix=
```

Block Limits Page:

```
    Peripheral Device Type: 0 (Direct Access Device)
    Peripheral Qualifier: 0 (Peripheral Device Connected)
        Page Code: 0xb0
        Page Length: 60 (0x3c)
        Write Same No Zero: False <- Note: We only support zero data!
Maximum Compare and Write Length: 1 block
Optimal Transfer Length Granularity: 0 blocks
    Maximum Transfer Length: 0 blocks
    Optimal Transfer Length: 0 blocks
Maximum Prefetch Transfer Length: 0 blocks
    Maximum Unmap LBA Count: 2097152 blocks (1073741824 bytes)
    Maximum Unmap Descriptor Count: 1
    Optimal Unmap Granularity: 8 blocks
    Unmap Granularity Alignment Valid: True
    Unmap Granularity Alignment: 0
    Maximum Write Same Length: 2097152 blocks (1073741824 bytes)
Maximum Atomic Transfer Length: 0
    Atomic Alignment: 0
Atomic Transfer Length Granularity: 0
Maximum Atomic Transfer Boundary: 0
Maximum Atomic Boundary Size: 0
```

```
#
```

Populate Disk Blocks

```
# spt cdb=8a starting=0 bs=256k emit=default pattern=iot slices=64 logprefix=
```

```
...
```

```
    Thread: 50
    Device Name: /dev/mapper/mpathu
    Device Info: Block Length=512, Capacity=409600000
    Block Range: 313600000 - 320000000
    SCSI Name: Write(16)
    SCSI CDB: 8a 00 00 00 00 00 13 12 ce 00 00 00 02 00 00 00
Data Direction: write
  Data Length: 262144
    Exit Status: 0 = SUCCESS
    Host Status: 0 = DID OK
    Driver Status: 0 = DRIVER OK
    SCSI Status: 0 = SCSI GOOD
    Sense Code: 0 = NO CODE
    Sense Key: 0 = NO SENSE
    asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 262144 (data bytes transferred)
    Residual: 0 (bytes not transferred)
    Iterations: 1
    Total Bytes: 3276800000
    Total Blocks: 6400000
Total Operations: 12501
    Sense Data: 00 00 00 00 00 00 00 00
  Elapsed Time: 01m44.75s
    Starting Time: Wed Dec 16 17:30:02 2020
    Ending Time: Wed Dec 16 17:31:47 2020
```

```
#
```

Report Mapped / Deallocated Blocks

```
# spt getlbastatus logprefix= emit=default
```

```
Get LBA Status Information:
```

```
    Mapped Blocks: 409600000
    Deallocated Blocks: 0
    Total Blocks: 409600000

    Thread: 1
    Device Name: /dev/mapper/mpathu
    Device Info: Block Length=512, Capacity=409600000
    Block Range: 0 - 409600000
    SCSI Name: Get LBA Status(16)
    SCSI CDB: 9e 12 00 00 00 00 18 60 00 00 00 00 28 a8 00 00
    Data Direction: read
    Data Length: 10408
    Exit Status: 0 = SUCCESS
    Host Status: 0 = DID OK
    Driver Status: 0 = DRIVER OK
    SCSI Status: 0 = SCSI GOOD
    Sense Code: 0 = NO CODE
    Sense Key: 0 = NO SENSE
    asc/asq: (00, 00) = No additional sense information
    Bytes Transferred: 24 (data bytes transferred)
    Residual: 10384 (bytes not transferred)
    Iterations: 1
    Total Bytes: 11544
    Total Blocks: 409600000
    Total Operations: 482
    Sense Data: 00 00 00 00 00 00 00 00 00 00
    Elapsed Time: 00m47.58s
    Starting Time: Wed Dec 16 16:07:20 2020
    Ending Time: Wed Dec 16 16:08:07 2020
```

<- FYI: Only one descriptor returned (spt requested 650 entries).

```
#
```

UNMAP Blocks

```
# spt unmap emit=default logprefix=
```

```
    Thread: 1
    Device Name: /dev/mapper/mpathu
    Device Info: Block Length=512, Capacity=409600000
    Block Range: 0 - 409600000
    SCSI Name: Unmap
    SCSI CDB: 42 00 00 00 00 00 00 00 18 00
Data Direction: write
  Data Length: 24
  Exit Status: 0 = SUCCESS
  Host Status: 0 = DID OK
  Driver Status: 0 = DRIVER OK
  SCSI Status: 0 = SCSI GOOD
  Sense Code: 0 = NO CODE
  Sense Key: 0 = NO SENSE
    asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 24 (data bytes transferred)
  Residual: 0 (bytes not transferred)
  Iterations: 1
  Total Bytes: 4704
  Total Blocks: 409600000
Total Operations: 198
  Sense Data: 00 00 00 00 00 00 00 00
Elapsed Time: 00m00.35s
  Starting Time: Wed Dec 16 16:10:33 2020
  Ending Time: Wed Dec 16 16:10:33 2020
```

```
#
```

Show Deallocated Blocks

```
# spt getlbastatus logprefix=
```

Get LBA Status Information:

```
Mapped Blocks: 0
Deallocated Blocks: 409600000
Total Blocks: 409600000
```

```
# spt getlbastatus logprefix= starting=1g/b limit=100g
```

Get LBA Status Information:

```
Mapped Blocks: 0
Deallocated Blocks: 209715200
Total Blocks: 209715200
```

```
#
```

Write Same w/UNMAP

```
# spt cdb=8a starting=0 bs=256k pattern=iot slices=64
# spt cdb=93,08 starting=0 emit=default logprefix=
```

```
      Thread: 1
      Device Name: /dev/mapper/mpathu
      Device Info: Block Length=512, Capacity=409600000
      Block Range: 0 - 409600000
      SCSI Name: Write Same(16)
      SCSI CDB: 93 08 00 00 00 00 18 60 00 00 00 0a 00 00 00 00
Data Direction: write
Data Length: 512
Exit Status: 0 = SUCCESS
Host Status: 0 = DID OK
Driver Status: 0 = DRIVER OK
SCSI Status: 0 = SCSI GOOD
Sense Code: 0 = NO CODE
Sense Key: 0 = NO SENSE
      asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 512 (data bytes transferred)
      Residual: 0 (bytes not transferred)
      Iterations: 1
      Total Bytes: 100352
      Total Blocks: 409600000
Total Operations: 197
      Sense Data: 00 00 00 00 00 00 00 00
      Elapsed Time: 00m00.35s
      Starting Time: Wed Dec 16 16:55:50 2020
      Ending Time: Wed Dec 16 16:55:50 2020
```

```
#
```

Zero ROD Method

```
# spt cdb=8a starting=0 bs=256k pattern=iot slices=64
# spt zerorod emit=default logprefix=
```

```
      Thread: 1
      Device Name: /dev/mapper/mpathu
      Device Info: Block Length=512, Capacity=409600000
      Block Range: 0 - 409600000
      SCSI Name: XCOPY - Write Using Token
      SCSI CDB: 83 11 00 00 00 00 01 8b 51 9b 00 00 02 28 00 00
Data Direction: write
  Data Length: 512
  Exit Status: 0 = SUCCESS
  Host Status: 0 = DID OK
  Driver Status: 0 = DRIVER OK
  SCSI Status: 0 = SCSI GOOD
  Sense Code: 0 = NO CODE
  Sense Key: 0 = NO SENSE
    asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 552 (data bytes transferred)
  Residual: 0 (bytes not transferred)
  Iterations: 1
  Total Bytes: 1725000
  Total Blocks: 409600000
Total Operations: 3127
  Sense Data: 00 00 00 00 00 00 00 00
Elapsed Time: 00m00.93s
  Starting Time: Wed Dec 16 17:02:44 2020
  Ending Time: Wed Dec 16 17:02:45 2020
```

```
#
```

dt I/O and UNMAP Commands

- dt UNMAP Options:
 - **unmap_freq=value** The SCSI unmap frequency. (Def: 0)
 - **unmap=type** The SCSI unmap type.
 - Valid types are: **random**, **unmap**, **write_same**, **zerorod**
- UNMAPs are done via *spt* commands (today)
- UNMAPs are *only* done between passes
- Options to control *spt*:
 - **spt_path=string** Path to SCSI (spt) tool.
 - Unix default: **/usr/local/bin/spt**
 - Windows default: **c:/tools/spt.exe**
 - **spt_options=string** Additional spt options.
- Roll Your Own via:
 - **pass_cmd=string** The per pass command to execute.
- BEWARE of **Linux BUG**: (external commands via **popen()** cause **false** data corruptions!)
 - Transient (read) corruption from linux host while running variable blocksize IO
 - Workaround: Use smaller block sizes (e.g. 256k) or limit to single thread (so no outstanding I/O).

DT Example with Random UNMAP Methods

```
# dt of=${DISK} bs=256k workload=high validation slices=5 passes=2 limit=1g unmap=random  
disable=get_lba_status 2>&1 | fgrep Executing
```

```
dt (j:1 t:1): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="93 08" starting=0 blocks=8192  
limit=419430b enable=sense,recovery logprefix= 2>&1
```

```
dt (j:1 t:2): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="83 11" starting=419430 limit=419430b  
enable=sense,recovery,zerorod logprefix= 2>&1
```

```
dt (j:1 t:3): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="93 08" starting=838860 blocks=8192  
limit=419430b enable=sense,recovery logprefix= 2>&1
```

```
dt (j:1 t:5): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="0x42" starting=1677720 limit=419432b  
enable=sense,recovery logprefix= 2>&1
```

```
dt (j:1 t:4): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="93 08" starting=1258290 blocks=8192  
limit=419430b enable=sense,recovery logprefix= 2>&1
```

```
dt (j:1 t:2): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="83 11" starting=419430 limit=419430b  
enable=sense,recovery,zerorod logprefix= 2>&1
```

```
dt (j:1 t:1): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="83 11" starting=0 limit=419430b  
enable=sense,recovery,zerorod logprefix= 2>&1
```

```
dt (j:1 t:4): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="93 08" starting=1258290 blocks=8192  
limit=419430b enable=sense,recovery logprefix= 2>&1
```

```
dt (j:1 t:3): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb=0x42 starting=838860 limit=419430b  
enable=sense,recovery logprefix= 2>&1
```

```
dt (j:1 t:5): Executing: /usr/local/bin/spt dsf=/dev/mapper/mpathu cdb="83 11" starting=1677720 limit=419432b  
enable=sense,recovery,zerorod logprefix= 2>&1
```

```
#
```

DT File Trim Operations

- File Systems with File Trim API's
 - Windows DeviceIoControl() **FSCTL_FILE_LEVEL_TRIM**
- dt File Trim Options:
 - **fstrim_freq=value** - The file system trim frequency (in **files**).
 - Flags:
 - **enable/disable=fstrim** - File system trim. (Default: **disabled**)
 - Like unmap options, applied *after* a full pass (between multiple passes)
- Example:

```
Administrator@rtp-fuji-qa32-1 /cygdrive/c/Robin
$ c:/tools/dt of=Y:/dt.data bs=random limit=25m workload=high_validation disable=scsi,stats passes=3
enable=fdebug,fstrim logprefix=
File name: Y:/dt.data
DEBUG: GetFileAttributesEx failed, error 2 - The system cannot find the file specified.
Directory: Y:, File: Y:/dt.data, Base Name: dt.data
Requesting file system information...
Trimming file Y:/dt.data, starting offset: 0, length: 26214400 bytes
Trimming file Y:/dt.data, starting offset: 0, length: 26214400 bytes
Trimming file Y:/dt.data, starting offset: 0, length: 26214400 bytes
Deleting file Y:/dt.data...

Administrator@rtp-fuji-qa32-1 /cygdrive/c/Robin
```

Debugging

```
# spt unmap logprefix= limit=3g enable=xdebug (debug adds full SCSI debug)
```

```
Range Descriptor: 0
Number of Blocks: 2097152 (0x200000)
Starting Logical Block Address: 0 (0x0) (lba's 0 - 2097151)
```

```
Range Descriptor: 0
Number of Blocks: 2097152 (0x200000)
Starting Logical Block Address: 2097152 (0x200000) (lba's 2097152 - 4194303)
```

```
Range Descriptor: 0
Number of Blocks: 2097152 (0x200000)
Starting Logical Block Address: 4194304 (0x400000) (lba's 4194304 - 6291455)
```

```
# spt eval 2097152b
```

```
Expression Values:
```

```
Decimal: 1073741824
Hexadecimal: 0x40000000
512 byte Blocks: 2097152.000000
Kilobytes: 1048576.000000
Megabytes: 1024.000000
Gigabytes: 1.000000
Terabytes: 0.000977
```

```
#
```

Futures

- Integrate *spt* into *dt* as an I/O Behavior
 - for example, **iobehavior=spt**
 - delayed (currently) due to effort required

Note: Not likely to happen now that I'm retired.



Backup Slides

- [FIO User Guide File Trim Options](#)
- FIO Options:
 - **trim_percentage=int** - Number of verify blocks to discard/trim.
 - **trim_verify_zero=bool** - Verify that trim/discarded blocks are returned as zeros.
 - **trim_backlog=int** - Trim after this number of blocks are written.
 - **trim_backlog_batch=int** - Trim this number of I/O blocks.